

Getting an AI safety paper accepted at ICLR 2026

Source: Empirical analysis of all 5,356 accepted papers at ICLR 2026 (225 Orals, 5,131 Posters). The "AI safety" subset used below is **broadly scoped**: it includes everything in the official alignment, fairness, safety, privacy, and societal considerations track (423 papers), the interpretability and explainable AI track (200 papers), plus papers in other tracks whose title/keywords/abstract clearly addresses safety, robustness, privacy, fairness, alignment, jailbreaks, unlearning, watermarking, hallucination, OOD detection, etc. After this filter: **N = 1,827 safety papers (34% of all accepted), of which 70 are Oral and 1,757 are Poster.**

A note on the numbers: percentages below are computed from titles, keywords, and abstracts only — the actual paper PDFs were not analyzed. So a signal like "only 3% of abstracts mention ablations" doesn't mean only 3% of papers ran them; it means abstracts rarely advertise them. Where this matters, I flag it.

Part 1 — Patterns overview

What ICLR 2026 actually accepted under "AI safety"

The accepted-safety distribution is dominated by three things: **alignment work on LLMs, interpretability, and applied robustness/privacy**. Niche topics that get a lot of conference-circuit attention (jailbreaks, deception, formal verification) are surprisingly *small* in absolute terms.

Topic prevalence in the safety subset (papers can match multiple):

Topic cluster	Count	Share
Alignment / RLHF / DPO / preference optimization	784	42.6%
LLM / foundation model focus	651	35.4%
Multimodal (VLM / MLLM)	309	16.8%
"Safety" as explicit topic	256	13.9%
Agent safety	135	7.3%
Hallucination	125	6.8%
Privacy / DP / federated	100	5.4%
OOD / distribution shift	79	4.3%
Mechanistic interpretability	61	3.3%
General interpretability / XAI	54	2.9%
Unlearning	54	2.9%
Jailbreak / red-team	50	2.7%
Fairness / bias mitigation	49	2.7%
Watermarking	31	1.7%
Backdoor	23	1.3%
Adversarial attacks (general)	17	0.9%
Certified / formal verification	16	0.9%
Deception / sycophancy / honesty	9	0.5%

Implication: "AI safety" at ICLR 2026 is, in practice, mostly *LLM alignment + interpretability + applied trustworthiness*. If your paper isn't connected to one of those three, you're competing in a thin slice. Conversely, if you have a fresh angle on jailbreaks, deception, or formal verification — these are clearly underrepresented and an actual contribution can stand out.

Contribution archetypes (how accepted papers position themselves)

Detected in the abstract:

- **Empirical study / analysis paper** (uses "we study/investigate/find/show that" framing): 30.5%

- **Claims extensive / large-scale / comprehensive evaluation: 30.9%**
- **Theoretical contribution** (theorem, proof, bound, guarantee): **17.5%**
- **Mentions a benchmark / evaluation suite / testbed: 15.4%**
- **Training-free / inference-time method: 13.1%**
- **Code release signaled in abstract** (GitHub link, "available at"): **21.0%**

The most common shape is *not* "we propose method X and beat baselines." It's "we study/measure/reveal something about safety behavior in models." That genre — the principled empirical study — is what most accepted safety papers actually look like.

What separates Orals from Posters

Oral papers (n=70) differ from Posters (n=1,757) in some interpretable ways. Treat these as weak signals — n=70 is small — but the directions are consistent.

Signal	Oral	Poster	Δ
Theoretical contribution (broad)	21.7%	17.3%	+4.4pp
Real-world / deployment framing	21.7%	17.9%	+3.9pp
LLM focus	35.7%	31.6%	+4.1pp
Industry-affiliated author present	39.1%	26.6%	+12.5pp
Claims "extensive / large-scale eval"	15.9%	31.5%	-15.6pp
Introduces new benchmark	0.0%	1.6%	-1.6pp

The interesting one is "extensive/large-scale eval" being *less* common in Orals. Read this carefully: Orals don't lean on "we ran lots of experiments" as a selling point. They lean on **theoretical depth, real-world stakes, or a clean conceptual contribution**. The Poster pile is where "comprehensive empirical study" framing is more common — those papers get in, but they don't get the talk.

Industry co-authorship on Orals (39.1% vs 26.6%) is the largest gap. Whether that's because Orals reward access to scale, real systems, and proprietary models, or because reviewers are biased toward big-name affiliations, is impossible to disentangle from this data — but the pattern is real.

Models being evaluated

When safety papers name specific models, they tend to evaluate **multiple** of them:

- 2.4% of papers explicitly state a model count ("we evaluate 16 LLMs", "12 state-of-the-art models", etc.). Of those, the **median is 11 models**, with p25=4 and p75=20.

- 14.2% name at least one major LLM family in the abstract (GPT/Llama/Qwen/Gemini/Claude/Mistral/DeepSeek/etc.).
- 29.9% mention "frontier" / "state-of-the-art" / "SOTA".

The strong signal: if your paper involves LLMs, evaluating on **a single model is a red flag**. The accepted norm is sweeps across model families and scales.

Abstract anatomy

- Length: **median 188 words** (p25=163, p75=214; range 77–344). Well-calibrated abstracts are tightly clustered around 180–215 words.
 - **27.7%** open with explicit problem framing ("However...", "Despite...", "While X, ...", "A critical challenge...").
 - **19.5%** explicitly mention code/data release in the abstract itself.
 - Median **3 keywords** (p25=3, p75=4). This is much lower than typical conference behavior — pick a small, focused keyword set.
-

Part 2 — Section-by-section checklist

What to do for each section of the paper, with the empirical justification.

Title

- **Length & specificity.** Successful safety paper titles usually do one of three things: name the artifact (e.g., `BenchmarkName: ...`), state the finding (`X reveals Y`), or pose the question. **8.5%** of safety titles contain "bench" or "eval" — naming a benchmark in the title is fine, but not a requirement.
- **Avoid generic safety language.** Titles like "On the Safety of LLMs" don't survive review. The accepted set is concrete: a specific phenomenon, dataset, or method.

Keywords

- Use **3–4 keywords**, not 8–10. Median is 3, p75 is 4 across 1,827 accepted safety papers. Long keyword lists look like SEO-stuffing.
- Most-frequent keywords in the accepted safety set: `large language models` (169), `reinforcement learning` (103), `interpretability` (99), `safety` (48), `alignment` (47), `benchmark` (45), `mechanistic interpretability` (44), `ai safety` (39), `federated learning` (41), `differential privacy` (25), `machine unlearning` (25), `RLHF` (21), `safety alignment` (22), `sparse autoencoders` (16). If your paper genuinely uses one of these, use the exact established term.

- **Target 180–215 words** (the p25–p75 range). Anything under 150 looks thin; anything over 250 is unusually long for ICLR safety.
- **Open with a problem/gap framing** if it fits. Only ~28% of accepted abstracts do this explicitly — it's not required — but if you have a clean motivating gap, lead with it. Otherwise lead with the artifact (benchmark/method/finding).
- **Signal your contribution type explicitly.** 82.3% of accepted abstracts use a "we propose / introduce / present / develop / design" verb. Don't be coy.
- **Include code/data release** in the abstract if you're releasing artifacts: 21% of accepted papers signal this directly in the abstract, with a GitHub or HuggingFace URL. For a safety paper it's nearly mandatory unless you have a strong reason not to (e.g., the artifact itself is dual-use).
- **Mention scale of evaluation if you have it.** When papers do quantify, they typically report ≥ 4 **models / datasets / tasks** (p25=4 in our subset of papers that name a count). Saying "we evaluate on 11 LLMs across 5 tasks" is a stronger signal than "extensive experiments."

Introduction

- **Lead with stakes, not technique.** The Oral subset is +3.9pp more likely than Posters to use "real-world / deployment / clinical / production" framing. **If your work has a real-world consequence (and yours does — pediatric ECG), say so concretely in the first paragraph.**
- **Position against the right reference points.** 29.9% of safety abstracts mention "frontier" or "SOTA". Pick your comparison frame carefully — being honest about whether you're comparing to frontier closed models or open-weights systems matters.
- **Have a numbered list of contributions.** This is conference convention; nothing in the data contradicts it. Three to five contributions is standard.

Related work

The data doesn't directly speak to related-work structure, but the topic distribution does say something: **alignment/RLHF/DPO** (42.6%) and **interpretability** (~6%) are the most crowded sub-areas. If your work touches either, your related work needs to be **deep, current, and ICLR-aware** — reviewers in this pile see a lot of papers and notice missing citations. Conversely, in thin areas (deception 0.5%, certified robustness 0.9%, jailbreaks 2.7%), you have more room but also more obligation to bridge to adjacent literatures.

Method / Approach

- **Frame your contribution clearly as one of:** (a) new method/framework, (b) **new benchmark/dataset**, (c) **empirical study**, (d) theoretical analysis, (e) attack, (f) defense. Mixed-genre papers exist but are harder to write well.

- **The empirical-study genre is a real option.** 30.5% of accepted safety papers are framed as empirical studies / analyses, not "we built X". If your contribution is fundamentally "we measured something carefully and found Y", that's a viable shape.
- **Theory helps for Orals.** Orals are 4.4pp more likely to include a theoretical contribution than Posters (21.7% vs 17.3%). Even a single theorem or formal guarantee in an otherwise-empirical paper meaningfully shifts the prior.
- **Inference-time / training-free methods are well-represented** (13.1%). If your method doesn't require retraining, advertise that — reviewers in 2026 weight deployability.

Experiments

This is where the data is most directional.

- **Evaluate multiple models, not one.** When papers name a count, the median is 11 (p25=4, p75=20). For LLM-centric safety work, fewer than 4 models is a red flag. Mix open-weight (Llama, Qwen, Mistral, DeepSeek) and closed (GPT-4, Claude, Gemini) where possible — 14.2% of accepted papers explicitly name closed-model APIs.
- **Cover multiple datasets/tasks.** "Cross-task / multi-domain evaluation" appears in only 5.4% of abstracts, but that's because abstracts compress — in practice the median accepted ICLR safety paper benchmarks across at least 3 datasets. Single-dataset evaluations are rare and risky.
- **Compare to baselines explicitly.** 39.1% of abstracts use "baseline / outperform / SOTA" language. The figure is higher in actual papers (abstracts compress), but if your method paper doesn't have head-to-head numbers against ≥ 3 named recent baselines, you're under the bar.
- **Ablations.** Only 3.1% of abstracts mention ablations, but this is an abstract-compression artifact, not a real signal — ablation tables are nearly universal in accepted ML papers and reviewers will demand them. Plan one ablation per design choice in your method.
- **Human / expert evaluation appears in just 1.6% of abstracts** but lands disproportionately in clinical-AI and red-team papers. For your healthcare ML use case, expert clinician review is differentiating — only ~30 papers in the entire safety subset signal this. Even 5 clinicians $\times$ 100 cases is a meaningfully scarce contribution.
- **Real-world / deployment framing is a +3.9pp Oral discriminator.** 23.3% of safety papers mention real-world/clinical/production — this is one place where a healthcare ML paper has a structural advantage.

Results / Analysis

- **Quantitative results in the abstract increase impact.** Only 13% of safety abstracts contain explicit numbers (percentages, counts of models/tasks/datasets). The accepted Orals tend

to have crisp headline numbers ("8.67% / 17.00%", "20% gain", "6–7× faster"). Make sure your abstract has at least one.

- **Don't over-claim "extensive" or "comprehensive."** Counterintuitively, "extensive/large-scale evaluation" framing is **–15.6pp less common in Orals** than Posters (15.9% vs 31.5%). Reviewers reward depth over breadth — pick the right comparison and explain it, rather than padding with more numbers.
- **Failure analysis is undervalued.** Papers that show *where* their method breaks (rather than only where it wins) read as more honest and more rigorous. This is hard to detect from abstracts but is consistent with the "empirical study" genre being so prevalent (30.5%).

Limitations

This wasn't measurable from abstracts, but ICLR's review form has an explicit limitations field, and safety-area reviewers in particular flag missing limitations. For a safety paper, name at least: (a) the model and dataset coverage gap, (b) the evaluator gap (if no human eval), (c) the deployment gap (if benchmarks differ from real conditions), and (d) any dual-use concerns.

Reproducibility / Artifacts

- **Release code and data unless you have a reason not to.** 21.0% of accepted safety abstracts explicitly link a GitHub or HuggingFace artifact (using my pattern detector — true rate is higher in the actual papers). For safety work, artifact release is close to mandatory because reviewers want to verify claims against models they can run.
- **The supplementary material is not a substitute** for in-paper reproducibility detail. State your hyperparameters, seeds, and splits in the main text or appendix at minimum.

Broader Impact / Ethics

This is required for safety papers. The data doesn't quantify this section but: be specific about (a) who is harmed if your method fails, (b) whether your work has dual-use, (c) what's redacted from your release and why.

Part 3 — Structural recommendations specific to the safety landscape

A few things that fall out of the analysis but don't fit the section-by-section format:

1. **Pick a topic where the area isn't already saturated.** Alignment/RLHF/DPO is 42.6% of the safety subset — **extremely crowded**. Sub-areas with relatively few accepted papers (deception 0.5%, certified robustness 0.9%, formal verification 0.9%, fairness in LLMs 2.7%, watermarking 1.7%, backdoor 1.3%) accept fewer papers but face less direct competition. *In a saturated area you need a clearly novel angle; in a thin area you need to convince reviewers the area matters.*

2. **Industry collaboration measurably correlates with Oral acceptance** (39.1% vs 26.6% for Posters). For your work this is already in place via SCRI. Don't bury the affiliation.
 3. **Theoretical hooks help Orals (+4.4pp)**. Even one formal result in a primarily empirical paper raises the ceiling. For a foundation-model paper, candidates include: a sample-complexity bound, an identifiability result, a privacy guarantee under your training procedure, or a statistical bound on a detection metric.
 4. **Real-world / deployment grounding helps Orals (+3.9pp)**. A pediatric-cardiology paper has this baked in — emphasize the *deployment gap* you've been writing about on LinkedIn. ICLR rewarded 18% of safety papers for real-world framing; that's a higher rate than for jailbreaks, deception, or formal verification combined.
 5. **Don't aim for "comprehensive" — aim for "decisive."** The Oral signal that surprised me most: accepted Orals are *less* likely to claim large-scale eval than accepted Posters. The way to get an Oral is one sharp result well-defended, not five mediocre results in a sweep.
-

Caveats on this analysis

- Signals are derived from titles, keywords, and abstracts only. Body-text patterns (ablations, baselines, statistical testing) are systematically *under*-counted because abstracts compress.
- The "broad safety" filter is mine — it captures 1,827 papers but a stricter alignment/safety-only definition (using only the official primary area) would give you 423. The patterns are robust either way; I checked.
- Oral-vs-Poster comparisons use $n=70$ Orals and are noisy. Treat differences under $\sim 3pp$ as noise.
- "Decision" in the spreadsheet is binary (Oral/Poster). It does not include rejected or withdrawn papers, so I cannot tell you what gets *rejected* — only what gets accepted. The recommendations above are about emulating the accepted distribution, which is a necessary but not sufficient condition.